

B.Tech. Degree VI Semester Special Supplementary Examination

January 2016

CS/IT 1602 COMPILER CONSTRUCTION

(2012 scheme)

Time : 3 Hours

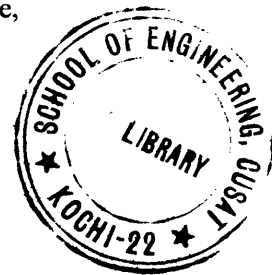
Maximum Marks : 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Define tokens, patterns and lexemes with examples.
- (b) Write a note on the lexical analyzer tool LEX.
- (c) Explain error recovery in operator precedence parsing.
- (d) Explain the configuration of an LR Parser on different moves like shift, reduce, accept and error.
- (e) Distinguish between synthesized and distributed attributes with examples.
- (f) What are activation records? Explain.
- (g) What is back patching?
- (h) Explain the different types of 3-address statements.



PART B

(4 × 15 = 60)

- II. Explain the phases in design of a compiler. (15)
- OR**
- III. What is the role of transition diagrams in the construction of a lexical analyser? (15)
Draw transition diagrams for (i) Relational operators (ii) Identifiers
- IV. Explain operator precedence parsing with example. (15)
- OR**
- V. Construct an SLR(1) parsing table for the following grammar. Also parse the string *float id, id* ; using that parse table. (15)

$$D \rightarrow \text{type tlist ;}$$

$$\text{type} \rightarrow \text{int} | \text{float}$$

$$\text{tlist} \rightarrow \text{tlist, id} | \text{id}$$
- VI. (a) Explain the various storage allocation strategies. (9)
(b) What do you mean by static and dynamic type checking? (6)
- OR**
- VII. With an example explain the bottom-up evaluation of S-attributed definition. (15)
- VIII. What are the principle sources of optimization in a code? Explain with the help of an example. (15)
- OR**
- IX. What are the different issues in the design of a code generator? (15)